Answer key – synthetic test set

Model risk management documents (JSC “Alazani Bank”): planted compliance gaps and their justification

Regulation	NBG Regulation on risk management of data-based statistical, AI and ML models (Order №151/04, 17.08.2020), with amendments №46/04 (02.05.2022) and №135/04 (26.06.2023)
Test documents	7 PDFs in Georgian, fictional commercial bank, dated 2021 – April 2022
Prepared	18.10.2026

1. How the test set works

All seven documents belong to one fictional commercial bank and are dated **before 2 May 2022**, so they reflect the regulation as originally adopted (V0). Each planted item has one of three classes:

- **A – amendment-triggered gap.** Compliant with V0 at the document date, but non-compliant once amendment 46/04 (V1) or 135/04 (V2) applies. These test whether the system detects what an amendment requires to be added or changed.
- **B – baseline non-compliance.** Already non-compliant with V0, the version in force at the document date. These test general compliance checking.
- **C – compliant control.** Sections that satisfy the requirement, sometimes in different wording. They must **not** be flagged, and measure false positives.

Version	Act	Applies	Source
V0	Order №151/04 (17.08.2020)	In force 01.01.2021; all models from 01.09.2021	matsne.gov.ge/ka/document/view/4964423?publication=0
V1	Amendment №46/04 (02.05.2022)	New or updated models immediately; all models from 01.10.2022	matsne.gov.ge/ka/document/view/5451815?publication=0
V2	Amendment №135/04 (26.06.2023)	From 01.07.2023	matsne.gov.ge/ka/document/view/5840572?publication=0

Commercial-bank note: several 135/04 changes (supervisory board reporting, board risk-appetite check, governance code for the third line) extend existing commercial-bank duties to microbanks. For a commercial bank, the only new substantive V2 duty planted here is the "must not hinder supervision" outsourcing condition (D1-09).

2. Summary by document

Doc	Title	Code / date	A	B	C
D1	Model Risk Management Policy D1_Model_Risk_Management_Policy_POL-RM-014.pdf	POL-RM-014 v1.0 15.03.2021	9	2	4
D2	Model Inventory D2_Model_Inventory_REG-RM-003.pdf	REG-RM-003 31.12.2021	1	4	0
D5	Internal Validation Report – M-001 D5_Validation_Report_M-001_VR-2021-07.pdf	VR-2021-07 25.11.2021	6	4	1

D6	Quarterly Model Monitoring Report Q1 2022 D6_Monitoring_Report_Q1-2022_MR-2022-Q1.pdf	MR-2022-Q1 28.04.2022	5	2	1
D9	AI Ethics Guideline D9_AI_Ethics_Guideline_EG-01.pdf	EG-01 v1.0 20.04.2021	8	0	1
D10	Internal Audit Report – MRM framework D10_Internal_Audit_Report_IA-2022-03.pdf	IA-2022-03 15.03.2022	3	5	1
D11	NBG consent application – M-002 D11_NBG_Consent_Application_M-002.pdf	NBG-APP-2021-02 10.08.2021	3	5	1
Total			**35**	**22**	**9**

3. Planted items with justification

D1 – Model Risk Management Policy (POL-RM-014 v1.0, 15.03.2021)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix
D1-01	A	5.2.1 (also 1.4.1)	All models get the same full annual validation with a single methodology, 'regardless of model type and purpose'. There is no risk-based model ranking (tiering).	**Art. 6 para 4¹–4²** – V1 – amendment 46/04 (02.05.2022) „41. შიდა ვალიდაციის პროცესი უნდა წარიმართოს რისკზე დაფუძნებული მიდგომის გამოყენებით.“ Validation must be risk-based: models must be ranked or grouped by materiality and model-use risk, and validation intensity must match the risk level.	Add a tiering methodology (materiality + use risk) and tier-dependent validation scope and frequency. Add a tier field to the inventory.
D1-02	A	5.3.2	Verification relies on the developer's documentation and tests plus sampled code review. Validators do not re-run the model with the first line's data and code.	**Art. 6 para 5 ბ.ა** – V1 – amendment 46/04 (02.05.2022) „ბ.ა) პროცესების ვერიფიკაციის ეტაპზე მოწმდება მოდელის კომპონენტების ფუნქციონირების გამართულობა:“ From 2022 the second line must run the model on the first line's data and code and compare the results, and must check data accuracy, code correctness and change control.	Require independent replication (re-run) by validators, with validator access to the development environment.

D1-03	A	4.6.1	Expert overrides are allowed whenever the owner 'considers' the score wrong. They are logged with a reason only, with no quantitative justification.	**Art. 6 para 5 ბ.ბ¹** – V1 – amendment 46/04 (02.05.2022) „ბ.ბ1) თუ მოდელის ვალიდაციის შედეგები არადამაკმაყოფილებელია და ახალი მონაცემებისა და…“ Replacing results or parameters by expert judgement must be justified by quantitative analysis, including back-testing, and kept under ongoing monitoring.	Add override governance: quantitative justification, back-testing of overrides, and monitoring thresholds.
D1-04	A	4.3.3	A model is escalated for restriction only when data accuracy is in doubt. The broader case where data quality standards cannot be confirmed is not covered, and restriction is optional ('will consider').	**Art. 5 para 5** – V1 – amendment 46/04 (02.05.2022) „5. თუ მოდელის გამომყენებელი სუბიექტი ვერ რწმუნდება, რომ მონაცემთა ხარისხთან…“ The 2022 wording makes failure to confirm that the data quality standards and policies are met grounds for restricting or suspending the model, not only doubts about accuracy.	Rewrite 4.3.3 so that non-confirmed data quality, measured against the 4.3.1 standard, triggers restriction or suspension.
D1-05	A	4.2.3, 4.4.2	Complex methods (gradient boosting, neural networks) are allowed purely for accuracy. Performance is assessed by Gini, KS and AUC only. There is no transparency or explainability requirement.	**Art. 5 para 10¹–10³** – V1 – amendment 46/04 (02.05.2022) „101. მოდელი უნდა იყოს გამჭვირვალე და უნდა იძლეოდეს ახსნისა და ინტერპრეტაციის შესაძლებლობას.“ Models must be explainable. Black-box models need global methods (PDP, variable importance, surrogate models) and local methods (ICE, Shapley values, LIME). The need grows with customer impact and automation.	Add an explainability section with required global and local methods for complex models and a transparency check during testing.
D1-06	A	2.1 (definitions), 9	There is no definition of 'deviation' (bias) and no requirement to avoid it. Ethics is limited to banning protected attributes as direct variables.	**Art. 2 ზ; Art. 9¹** – V1 – amendment 46/04 (02.05.2022) „ზ) გადახრა – მონაცემებში ან/და ალგორითმში არსებული კანონზომიერება, რომლის გამოც სისტემატურად…“ The 2022 amendment adds a definition of deviation and a new Article 9¹: avoiding deviation and ensuring transparency are minimum principles, with mitigation measures a–g.	Add the deviation definition and an ethics chapter covering Art. 9¹ (principles, deviation types, mitigation measures, customer explanation and awareness).

D1-07	A	9.1	It cites the 2018 ethics code (Order №154/04 of 09.07.2018).	**Art. 4 para 3** – V1 – amendment 46/04 (02.05.2022) „3. მოდელის მართვის პროცესში მოდელის გამომყენებელმა სუბიექტმა უნდა შეიმუშაოს ეთიკის პრინციპები…“ From 2022 commercial banks must follow the ethics code approved by NBG Order №13/04 of 02.02.2021. The 154/04 reference is outdated.	Replace the reference with Order №13/04 (02.02.2021).
D1-08	A	7.1.2–7.1.3	Third-party controls cover only documentation at purchase. There is no periodic receipt of the vendor's test results, no in-house monitoring capability if the vendor does not monitor, no own-data testing when vendor documents are missing, and no risk-based review.	**Art. 9 para 4, 4¹–4³** – V1 – amendment 46/04 (02.05.2022) „41. მოდელის გამომყენებელი სუბიექტი მესამე მხარისგან რეგულარულად უნდა იღებდეს…“ Since 2022: test on own data if vendor documents are unavailable; regularly obtain the vendor's test results and check them against risk appetite; have resources to monitor if the vendor doesn't; review purchased models on a risk basis.	Extend 7.1 with the 9.4–4³ obligations and the matching contractual clauses (SLA).
D1-09	A	7.2.1	The outsourcing clause refers to the 47/04 principles and three-lines independence, but has no condition that outsourcing must not hinder supervision.	**Art. 9 para 1** – V2 – amendment 135/04 (26.06.2023) „რათა ხელი არ შეეშალოს ზედამხედველობის პროცესს“ The 2023 amendment adds that outsourcing of model building, validation or audit must not hinder the supervisory process.	Add the 'must not hinder supervision' condition, for example NBG access rights in outsourcing contracts.
D1-10	B	3.3.1–3.3.2, 4.5.1	Launch is approved by the Model Committee, which is chaired by the CRO (second line) and includes the head of RAMD (developer and validator). The approver is not independent of the lines of defence.	**Art. 8 para 1** – V0 – Order 151/04 (17.08.2020) „თანხმობა უნდა გასცეს დაცვის ხაზებისაგან დამოუკიდებელმა პირმა/ორგანომ“ Approval to launch must come from a person or body independent of the lines of defence, with technical competence.	Assign approval to a body or person independent of the lines of defence, such as a management-level committee without the CRO and RAMD voting, or the CEO with advice.

D1-11	B	3.5.1–3.5.2	The validation group sits inside RAMD with the development groups, and the head of RAMD approves validation reports. Independence relies only on 'not validating your own model'.	**Art. 6 para 4; Art. 4 para 1** – V0 – Order 151/04 (17.08.2020) „შიდა ვალიდაცია მოდელის აგებისა და გამოყენებისგან დამოუკიდებლად უნდა განხორციელდეს.“ Validation must be independent of development and use. The three lines must be independent.	Separate the validation function from model development, with a separate reporting line, for example to the CRO directly or a separate unit.
D1-C1	C	3.4.1	Three lines described as 'business function / control function / independent assurance', with the second line's power to restrict or ban use.	**Art. 4 para 1** – V0 – Order 151/04 (17.08.2020) „მას უნდა გააჩნდეს უფლებამოსილება შეზღუდოს ან აკრძალოს მოდელის გამოყენება“ Compliant control in different wording. It should NOT be flagged.	—
D1-C2	C	7.3.1	A contingency plan is required for every third-party model.	**Art. 9 para 5** – V0 – Order 151/04 (17.08.2020) „გეგმა იმ შემთხვევისთვის, თუ მესამე მხარის მოდელი აღარ იქნება ხელმისაწვდომი“ Compliant. It should NOT be flagged. (The inventory D2 violates this policy rule; see D2-02.)	—
D1-C3	C	4.2.2, 4.4.1	Proxy variables are documented, and scenario testing (baseline, optimistic, pessimistic) is required.	**Art. 5 para 8, 10** – V0 – Order 151/04 (17.08.2020) „სხვადასხვა საბაზრო სიტუაციის პირობებში (საბაზისო, ოპტიმისტური, პესიმისტური)“ Compliant. It should NOT be flagged.	—
D1-C4	C	8.1–8.3	Supervisory models: NBG consent, monthly reporting, annual update, prior notice of changes.	**Art. 10–11** – V0 – Order 151/04 (17.08.2020) „ყოველთვიურად წარუდგინოს საქართველოს ეროვნულ ბანკს ანგარიშგება“ Compliant at policy level. It should NOT be flagged. (D11 contradicts it; see D11-02 and D11-03.)	—

D2 – Model Inventory (REG-RM-003, 31.12.2021)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix

D2-01	A	Section 1 note; 'next validation' column	There is no risk tier or materiality attribute. Every model's next validation is set mechanically at 12 months.	**Art. 6 para 4¹–4²** – V1 – amendment 46/04 (02.05.2022) „42. მოდელის რისკის დონის მიხედვით მოდელის გამომყენებელმა სუბიექტმა უნდა განსაზღვროს…“ Validation must be risk-based, which needs a model ranking or grouping by risk level.	Add tier, materiality and use-risk columns; derive validation frequency from the tier.
D2-02	B	Row M-005 – 'contingency plan' column	The vendor fraud model M-005 has 'no contingency plan'.	**Art. 9 para 5** – V0 – Order 151/04 (17.08.2020) „გეგმა იმ შემთხვევისთვის, თუ მესამე მხარის მოდელი აღარ იქნება ხელმისაწვდომი“ A plan is required in case a third-party model becomes unavailable. It also breaks the bank's own policy, 7.3.1.	Define and document a contingency plan for M-005.
D2-03	B	Row M-007 – 'last validation' column	The vendor collections model M-007 is in production with no internal validation; the vendor's report was accepted instead.	**Art. 6 para 3; Art. 3 para 3; Art. 8 para 1** – V0 – Order 151/04 (17.08.2020) „შესყიდული მოდელის შიდა და მესამე მხარის მიერ შესასრულებელი ვალიდაციისათვის აუცილებელი მოთხოვნები“ Internal validation covers third-party models too, and launch requires a positive assessment by all lines.	Perform the internal validation of M-007; restrict use until it is complete.
D2-04	B	Row M-002	The supervisory income model M-002 has been in production since 01.06.2021. NBG consent was only applied for on 10.08.2021 and is still pending, and there is no internal validation.	**Art. 10–11; Art. 8 para 1** – V0 – Order 151/04 (17.08.2020) „2020 წლის 13 მარტის №44/04 ბრძანებით დამტკიცებული დებულებით“ The income assessment model is a supervisory model whose use requires NBG's prior consent, and an internal validation must be part of the application.	Stop or limit use until consent is granted; complete the internal validation.
D2-05	B	Row M-005 – 'last validation' column	The 'validation' of M-005 was only a review of the vendor's documentation.	**Art. 6 para 3** – V0 – Order 151/04 (17.08.2020) Validation must cover inputs, calculations and outputs, for in-house and third-party models alike. A documentation review alone is partial.	Perform a full validation of M-005, or record it explicitly as partial with a remediation plan.

D5 – Internal Validation Report – M-001 (VR-2021-07, 25.11.2021)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix
D5-01	A	5.1–5.2	Validators did not re-run the model (no access to the development environment) and relied on the developer's own LOS-versus-model comparison.	**Art. 6 para 5 ბ.ა** – V1 – amendment 46/04 (02.05.2022) „ბ.ა) პროცესების ვერიფიკაციის ეტაპზე მოწმდება მოდელის კომპონენტების ფუნქციონირების გამართულობა:“ The second line must run the model with the first line's data and code and compare the results.	Give validators access to the environment; replicate the scores independently.
D5-02	A	7.1–7.2	The only interpretability evidence is LightGBM gain importance, and further analysis is called 'not needed'. Yet 78% of decisions are automated and customer-facing.	**Art. 5 para 10²–10³** – V1 – amendment 46/04 (02.05.2022) „102. იმ ტიპის მოდელების შემთხვევაში, სადაც რთულია სრული წარმოდგენის შექმნა…“ Black-box models require global methods (PDP, variable importance, surrogate tree) and local methods (ICE, SHAP, LIME). The need increases with customer impact and automation.	Add PDP/ICE, SHAP and surrogate-model analysis; assess whether reason codes can be produced for customers.
D5-03	A	2.2–2.3, 7.1	Municipality (as an income proxy) and age are among the top-5 drivers. There is no bias or deviation testing and no check that the proxy has a causal link to the outcome.	**Art. 9¹ para 3–4 ე** – V1 – amendment 46/04 (02.05.2022) „ჩამნაცვლებელი ცვლადი არ უნდა გამოიყენებოდეს…“ Proxy variables must not be used unless they have a causal link to the outcome. Deviation must be identified and mitigated.	Run fairness and deviation analysis (by age, region); justify causality or remove the proxy.
D5-04	A	3.1, 3.3	The sample covers 2015–2020, including the 2020 deferral period, with no assessment of outdated data, and this was 'not considered material'.	**Art. 9¹ para 4 დ** – V1 – amendment 46/04 (02.05.2022) „დ) მოძველებული მონაცემების გამოყენების კონტროლი და აღმოფხვრა, თუ მონაცემებში არსებული ტენდენცია…“ Use of outdated data that no longer reflects current conditions must be controlled and removed.	Test time-stability, exclude or weight distorted periods, and document the decision.

D5-05	A	1.1	The scope is the 'standard scope applied equally to all models', not linked to model risk.	**Art. 6 para 4²** – V1 – amendment 46/04 (02.05.2022) „42. მოდელის რისკის დონის მიხედვით მოდელის გამომყენებელმა სუბიექტმა უნდა განსაზღვროს…“ Validation intensity must correspond to the model's risk level.	State the model's tier and justify the scope.
D5-06	A	8.2 and recommendation R2	The validators recommend cutting the cut-off by 40 points for one segment based on expert opinion, without re-estimation or quantitative analysis.	**Art. 6 para 5 ბ.ბ¹** – V1 – amendment 46/04 (02.05.2022) „ბ.ბ1) თუ მოდელის ვალიდაციის შედეგები არადამაკმაყოფილებელია და ახალი მონაცემებისა და…“ Expert-judgement replacement of parameters must be justified quantitatively, including back-testing, and monitored.	Base the cut-off change on quantitative analysis and back-testing; define monitoring.
D5-07	B	4.3	Sensitivity and stability tests under extreme values were not performed, on the assumption that tree models are robust.	**Art. 6 para 5 ა** – V0 – Order 151/04 (17.08.2020) The conceptual soundness assessment must include sensitivity and stability tests, including extreme values.	Perform sensitivity and stress tests on the inputs.
D5-08	B	Section 10 (signatures); D1 3.5	The report is approved by the head of RAMD, who also supervises the scoring (development) group. The validator sits in the same department.	**Art. 6 para 4** – V0 – Order 151/04 (17.08.2020) „შიდა ვალიდაცია მოდელის აგებისა და გამოყენებისგან დამოუკიდებლად უნდა განხორციელდეს.“ Validation must be independent of development.	Have the report approved by an independent validation head who reports outside development.
D5-09	B	1.2	The model was launched (01.07.2021) on a temporary permission before validation was completed.	**Art. 8 para 1** – V0 – Order 151/04 (17.08.2020) „თანხმობა უნდა გასცეს დაცვის ხაზებისაგან დამოუკიდებელმა პირმა/ორგანომ“ Launch is allowed only after all three lines have positively assessed the model and an independent approver has agreed.	Do not launch before validation; if urgent, restrict use and document conditions.
D5-10	B	9.1 vs 6.2	Rated 'satisfactory, no restrictions' even though back-testing was impossible and several points above are open.	**Art. 6 para 6; Art. 8 para 1–3** – V0 – Order 151/04 (17.08.2020) „შეიმუშავებს საბოლოო დასკვნას მოდელის რისკის შესახებ“ The final conclusion must reflect model risk; unresolved gaps should lead to restrictions.	Downgrade the rating or attach use restrictions until the gaps are closed.

D5-C1	C	3.2, 6 (benchmark)	Data quality checks and the benchmark comparison against the logistic scorecard are performed.	**Art. 5 para 4; Art. 6 para 5 ბ.ფ** – V0 – Order 151/04 (17.08.2020) „მოდელის შედეგებსა და საორიენტაციო მოდელს შორის განსხვავების არსებობის დროს“ Compliant. It should NOT be flagged.	—

D6 – Quarterly Model Monitoring Report Q1 2022 (MR-2022-Q1, 28.04.2022)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix
D6-01	B	Section 7	Distribution goes to the CRO, Model Committee and Management Board only, not to the Supervisory Board (Risk Committee). That also breaks the bank's own policy, 6.1.	**Art. 6 para 5 ბ.ე** – V0 – Order 151/04 (17.08.2020) „კომერციული ბანკის შემთხვევაში – სამეთვალყურეო საბჭოსთან“ For commercial banks, regular model risk reporting must also go to the supervisory board.	Add the Supervisory Board Risk Committee to the distribution.
D6-02	B	2.2–2.3	M-001's PSI = 0.27 (red) and the digital-channel actual/predicted ratio = 1.46, yet the model 'remains in use without restriction'. There is no escalation and no restriction.	**Art. 8 para 3** – V0 – Order 151/04 (17.08.2020) „უნდა გახდეს მოდელის გამოყენების შეზღუდვის ან შეჩერების საფუძველი“ A significant change in model risk is grounds for restriction or suspension, and must be reported to the approver. The policy's 6.3 requires immediate escalation.	Escalate to the approver; consider restricting automated approvals for the digital channel; bring re-calibration forward.
D6-03	A	2.2.3	The 40-point cut-off change, based on expert opinion, was implemented. Success is measured only by fewer overrides, with no back-testing of risk outcomes.	**Art. 6 para 5 ბ.ბ¹** – V1 – amendment 46/04 (02.05.2022) „ბ.ბ1) თუ მოდელის ვალიდაციის შედეგები არადამაკმაყოფილებელია და ახალი მონაცემებისა და…“ Expert-judgement replacement needs quantitative justification, including back-testing, and monitoring.	Back-test the default rate of the affected segment; report it quantitatively.

D6-04	A	3.2	M-003 has 12.4% missing values (above the 2% threshold). They were imputed with the mode, a +3% overlay was applied, and the model is used without restriction.	**Art. 5 para 5; Art. 8 para 3** – V1 – amendment 46/04 (02.05.2022) „5. თუ მოდელის გამომყენებელი სუბიექტი ვერ რწმუნდება, რომ მონაცემთა ხარისხთან…“ Failure to confirm that data quality standards are met must be grounds for restricting or suspending the model.	Formally restrict or condition use, with a documented decision by the approver, until the data is restored.
D6-05	A	4.1–4.3	No periodic vendor test results for M-005, no contractual right to them, and no internal means of monitoring. The model stays in use anyway.	**Art. 9 para 4¹–4²** – V1 – amendment 46/04 (02.05.2022) „42. თუ შეძენილი მოდელის მუდმივი მონიტორინგი მესამე მხარის მიერ არ ხორციელდება…“ The bank must regularly receive the vendor's test results, and must have its own resources to verify performance if the vendor doesn't monitor.	Amend the SLA; obtain score-level data; build internal monitoring.
D6-06	A	Whole report (no section)	There is no deviation or bias monitoring, even though the digital-channel shift involves younger applicants and the model uses age.	**Art. 9¹ para 4 გ** – V1 – amendment 46/04 (02.05.2022) Ongoing monitoring must check for deviation.	Add deviation metrics (approval and default rates by age group and region) to the monitoring report.
D6-07	A	Annex A, row M-007	The vendor model M-007 is rated 'green' solely on the vendor's own report.	**Art. 9 para 4³** – V1 – amendment 46/04 (02.05.2022) „43. მოდელის გამომყენებელმა სუბიექტმა რისკზე დაფუძნებული მიდგომით უნდა განახორციელოს…“ Purchased models must be reviewed by the bank on a risk basis.	Perform an independent risk-based review of M-007.
D6-C1	C	Section 5	The override analysis is performed and reported.	**Art. 6 para 5 ბ.გ** – V0 – Order 151/04 (17.08.2020) „მოდელის შედეგების ექსპერტული მსჯელობით ჩანაცვლების შემთხვევები უნდა იყოს გაანალიზებული“ Compliant. It should NOT be flagged.	—

D9 – AI Ethics Guideline (EG-01 v1.0, 20.04.2021)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix

D9-01	A	Section 2, 2nd bullet	It cites the 2018 ethics code (№154/04).	**Art. 4 para 3** – V1 – amendment 46/04 (02.05.2022) „3. მოდელის მართვის პროცესში მოდელის გამომყენებელმა სუბიექტმა უნდა შეიმუშაოს ეთიკის პრინციპები…“ Commercial banks must follow the code approved by NBG Order №13/04 (02.02.2021).	Update the reference to 13/04.
D9-02	A	Section 3 (principles)	The core principles leave out avoiding deviation and transparency. There is no definition of deviation and none of the four deviation types.	**Art. 9¹ para 1–3; Art. 2 ზ** – V1 – amendment 46/04 (02.05.2022) „1. მოდელის განვითარების, დანერგვისა და გამოყენების პროცესში მოდელის გამომყენებელმა სუბიექტმა უნდა დაიცვას…“ At minimum, avoiding deviation and ensuring transparency are required principles. Selection, stereotypical, behavioural and exclusion deviation must be addressed.	Add the principles, the definition and the deviation typology.
D9-03	A	4.2	Geographic and demographic variables are allowed whenever they improve predictive power, which is effectively a proxy licence with no causal test.	**Art. 9¹ para 4 ე** – V1 – amendment 46/04 (02.05.2022) „ჩამნაცვლებელი ცვლადი არ უნდა გამოიყენებოდეს…“ Proxy variables must not be used unless they have a causal link to the outcome. All relevant customer micro-characteristics should be integrated instead.	Replace 4.2 with a causal-justification test and a fairness review for proxies.
D9-04	A	3.2	Accuracy is declared the 'primary objective' and the main selection criterion.	**Art. 9¹ para 4 ვ** – V1 – amendment 46/04 (02.05.2022) Socially neutral outcomes should be prioritised if this does not significantly reduce accuracy.	Rewrite the principle to balance accuracy with neutral outcomes.
D9-05	A	5.1–5.2	Model logic is a trade secret, and refusal reasons are never explained to customers.	**Art. 9¹ para 5; Art. 5 para 10³** – V1 – amendment 46/04 (02.05.2022) „5. მოდელის გამომყენებელმა სუბიექტმა უნდა უზრუნველყოს მოდელის გამოყენებით მიღებული გადაწყვეტილების ლოგიკის გამჭირვალობის ზრდა.“ The bank must increase the transparency of model-based decision logic and be able to explain it to customers when needed.	Introduce customer-level explanations (reason codes) while protecting proprietary detail.

D9-06	A	Whole document (missing)	There are no customer education or awareness measures about AI and machine-learning decisions.	**Art. 9¹ para 6** – V1 – amendment 46/04 (02.05.2022) „6. მოდელის გამომყენებელმა სუბიექტმა ხელი უნდა შეუწყოს მანქანური სწავლებისა და ხელოვნური ინტელექტის…“ The bank must promote customer education and awareness about decisions made with machine learning and AI.	Add customer communication and awareness measures, for example a website notice and information in loan documents.
D9-07	A	Section 7 (roles)	No procedures to identify, report and mitigate deviation in data processing and ongoing monitoring; the validators' role is limited to checking the banned-variable list. There's also no control of outdated data.	**Art. 9¹ para 4 ა–დ, ზ** – V1 – amendment 46/04 (02.05.2022) „დ) მოძველებული მონაცემების გამოყენების კონტროლი და აღმოფხვრა, თუ მონაცემებში არსებული ტენდენცია…“ The bank must set rules and procedures to identify, report and mitigate deviation; monitor it in data processing and ongoing monitoring; remove outdated data; and control inappropriate algorithms.	Add a deviation-management procedure with roles across all three lines.
D9-08	A	Whole document (missing)	There is no explainability requirement for complex or black-box models.	**Art. 5 para 10¹–10²** – V1 – amendment 46/04 (02.05.2022) „10¹. მოდელი უნდა იყოს გამჭვირვალე და უნდა იძლეოდეს ახსნისა და ინტერპრეტაციის შესაძლებლობას.“ Models must be explainable, with global and local methods for black boxes.	Add explainability requirements, or cross-reference the updated policy.
D9-C1	C	6.1–6.3, 3.5	The personal data protection and human-review provisions are present.	**—** – V0 – Order 151/04 (17.08.2020) Compliant or neutral. It should NOT be flagged as non-compliant with 151/04.	—

D10 – Internal Audit Report – MRM framework (IA-2022-03, 15.03.2022)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix

D10-01	B	2.2 (bullets 1–2)	Audit scope excludes the documentation of model updates and the operational systems and data management, which are deferred to 2023.	**Art. 7 para 2** – V0 – Order 151/04 (17.08.2020) „2. დაცვის მესამე ხაზი უნდა დარწმუნდეს, რომ მოდელის აგების, შიდა ვალიდაციისა და განახლების…“ The third line must confirm that model building, validation and updates are documented, and must assess the operational systems and data management. This is in V0 and restated in V1.	Bring update documentation and systems/data management into scope.
D10-02	B	2.2 (bullet 3)	The timeliness of validation was not assessed.	**Art. 7 para 1** – V0 – Order 151/04 (17.08.2020) The audit must confirm that validation is completed on time.	Assess validation timeliness against the plan and the inventory.
D10-03	B	3.2	RAMD analysts who built M-001 helped with the audit's recalculations and data extraction.	**Art. 4 para 1; Art. 7** – V0 – Order 151/04 (17.08.2020) The three lines of defence must be independent, so the third line can't rely on the developers to test their own model.	Use independent data access or external specialists; document safeguards.
D10-04	B	3.3	The co-source audit partner previously advised on the M-003 methodology.	**Art. 9 para 1** – V0 – Order 151/04 (17.08.2020) „2014 წლის 13 ივნისის №47/04 ბრძანებით“ Outsourced audit must follow the 47/04 principles and preserve the independence of the three lines.	Assess conflicts of interest; exclude M-003 from the partner's work or replace the partner.
D10-05	A	3.1	The methodology refers only to the IIA standards, not the Corporate Governance Code for commercial banks (Order 215/04).	**Art. 7 para 3** – V1 – amendment 46/04 (02.05.2022) „3. კომერციული ბანკის შემთხვევაში, დაცვის მესამე ხაზი ასევე უნდა აკმაყოფილებდეს…“ From 2022 the third line at a commercial bank must also meet the Corporate Governance Code, Order 215/04 of 26.09.2018.	Add the 215/04 requirements to the audit methodology.
D10-06	A	Finding F6	The audit concludes that having no tiering 'complies with the regulation' and makes no recommendation.	**Art. 6 para 4¹–4²** – V1 – amendment 46/04 (02.05.2022) „41. შიდა ვალიდაციის პროცესი უნდა წარიმართოს რისკზე დაფუძნებული მიდგომის გამოყენებით.“ This was correct under V0, but from 2022 a risk-based tiering methodology is mandatory, so the conclusion becomes wrong.	Re-rate F6 as a finding with a recommendation.

D10-07	A	Section 2.1 (scope) – missing	The scope has no test of compliance with the ethics, deviation and explainability requirements.	**Art. 7 para 1; Art. 9¹; Art. 5 para 10¹** – V1 – amendment 46/04 (02.05.2022) „101. მოდელი უნდა იყოს გამჭვირვალე და უნდა იძლეოდეს ახსნისა და ინტერპრეტაციის შესაძლებლობას.“ The audit must confirm compliance with the regulation, which from 2022 includes Art. 9¹ and 5.10¹–10³.	Add ethics, deviation and explainability tests to the audit programme.
D10-08	B	F3, F4 ratings; 5.2	A model launched before validation (F3) and a supervisory model used without NBG consent (F4) are rated only 'medium', and 5.2 says the processes 'comply'.	**Art. 8 para 1; Art. 10–11** – V0 – Order 151/04 (17.08.2020) „თანხმობა უნდა გასცეს დაცვის ხაზებისაგან დამოუკიდებელმა პირმა/ორგანომ“ These are direct breaches of the regulation, so the ratings and conclusion understate the risk.	Re-rate F3 and F4 as high; correct the 5.2 conclusion.
D10-C1	C	F1, F2	Findings on the missing M-007 validation and the incomplete inventory.	**Art. 6 para 3; Art. 3** – V0 – Order 151/04 (17.08.2020) Correct findings. They should NOT be flagged as wrong.	—

D11 – NBG consent application – M-002 (NBG-APP-2021-02, 10.08.2021)

ID	Class	Where in document	What the document says	Requirement (article, version, quote)	Expected fix
D11-01	B	Section 6	No internal validation results; validation is promised 'after NBG consent'.	**Art. 11 para 2** – V0 – Order 151/04 (17.08.2020) The application must include the internal validation results. It also breaks the bank's own policy, 8.2.	Complete the internal validation and attach the report.
D11-02	B	12.1	Reporting to the NBG is offered quarterly.	**Art. 11 para 5** – V0 – Order 151/04 (17.08.2020) „ყოველთვიურად წარუდგინოს საქართველოს ეროვნულ ბანკს ანგარიშგება“ A monthly report on the supervisory model is required.	Commit to monthly reporting.
D11-03	B	12.2 (1st sentence)	The model update is planned every 2 years.	**Art. 11 para 5** – V0 – Order 151/04 (17.08.2020) „განაახლოს საზედამხედველო მოდელი მინიმუმ ყოველწლიურად“ The supervisory model must be updated at least annually.	Commit to an update at least annually.

D11-04	B	12.2 (2nd sentence)	Significant changes would be notified within 30 days after implementation.	**Art. 11 para 5** – V0 – Order 151/04 (17.08.2020) Significant changes must be disclosed and justified with error analysis, and approved by the NBG within 30 working days. Until approval, the pre-update model stays in use.	Change to prior notification and approval.
D11-05	B	Section 2	The model has been used in the credit process since 01.06.2021, before the application date of 10.08.2021.	**Art. 10; Art. 11** – V0 – Order 151/04 (17.08.2020) „2020 წლის 13 მარტის №44/04 ბრძანებით დამტკიცებული დებულებით“ The income assessment model is a supervisory model; its use requires prior NBG consent.	Suspend or limit its use, or explain the transitional arrangement, pending consent.
D11-06	A	5.1, 8.2	The Random Forest component is declared 'not directly interpretable', with no global or local explainability, even though decisions are automated without expert involvement.	**Art. 5 para 10^{2}–10^{3}** – V1 – amendment 46/04 (02.05.2022) „103. გამჭირვალობის საჭიროება იზრდება, თუ მოდელის გამოყენებით მიღებული გადაწყვეტილებები…“ Black-box components need explainability methods, and the need is higher when decisions are automated and affect customers.	Add SHAP, PDP and surrogate analysis for the Random Forest component.
D11-07	A	4, 5.2, 10.1	Municipality and age are used, and the benchmark matrix is region-based. There is no deviation or bias analysis for income estimation.	**Art. 9^{1} para 3–4** – V1 – amendment 46/04 (02.05.2022) „ჩამნაცვლებელი ცვლადი არ უნდა გამოიყენებოდეს…“ Deviation must be avoided, and proxies are allowed only with a causal link.	Add a deviation analysis by region and age; justify or remove the proxies.
D11-08	A	4 (data 2017–2020)	Training data spans 2017–2020, including the pandemic, with no outdated-data assessment.	**Art. 9^{1} para 4 დ** – V1 – amendment 46/04 (02.05.2022) „დ) მოძველებული მონაცემების გამოყენების კონტროლი და აღმოფხვრა, თუ მონაცემებში არსებული ტენდენცია…“ Outdated data that no longer matches current conditions must be controlled and removed.	Document a time-stability analysis and data window selection.
D11-C1	C	1–5, 9, 10	Owner, responsible persons, creation date, purpose, segment, data sources, construction, assumptions, limitations, expert alignment and benchmark are all included.	**Art. 11 para 2** – V0 – Order 151/04 (17.08.2020) Compliant parts of the application. They should NOT be flagged.	—

4. Sources

- Order №151/04, as originally adopted: https://www.matsne.gov.ge/ka/document/view/4964423?publication=0
- Amendment №46/04 (02.05.2022): https://www.matsne.gov.ge/ka/document/view/5451815?publication=0
- Amendment №135/04 (26.06.2023): https://www.matsne.gov.ge/ka/document/view/5840572?publication=0
- Structural reference for typical document types and contents (no Georgian public samples exist): PRA SS1/23 Model risk management principles for banks (prarulebook.co.uk); US interagency SR 11-7 guidance as republished by the FDIC (fdic.gov/sites/default/files/2024-03/fil17022a.pdf).

5. Limitations

- The Georgian quotes are short verbatim openings of the cited paragraphs, taken from the Matsne pages. Superscript paragraph numbers appear on Matsne as plain digits (e.g. “41.” = 4¹). Check paragraph labels against the official PDF before using them as gold labels.
- Where one quote covers several related duties (e.g. Art. 11 para 5), the quote is the opening of that paragraph and the explanation names the specific duty.
- Some items have a mixed basis. For example, D6-04 is primarily the V1 wording of Art. 5 para 5, but Art. 8 para 3 (V0) also supports it. The class records the primary basis.
- The bank, people, vendors and figures are fictional. The documents are deliberately imperfect test material, not templates for real use.